

Class Meeting Handout #9

Barry Linkletter

We often see reactions that are catalyzed by acid or base, especially in biological mechanisms. Today we will begin an exploration of acid and base catalysis.

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*,

Acid & Base Catalysis

Understanding the nature of acid and base catalysis is the first step in understanding the mechanism of enzyme catalysis and many other important chemical processes.¹

¹ Read chapter 9 of the textbook.

Discussion: The Definitions – 10 minutes

We must first agree on our words.

1. Describe as many forms of chemical catalysis as you can.²
2. We will be concentrating entirely on acid and base catalysis in this part of the course. So please state the following definitions...
 - (a) Define what is meant by acid catalysis.
 - (b) Define what is meant by base catalysis.
 - (c) Define what is meant by specific catalysis.
 - (d) Define what is meant by general catalysis.

² Ch. 9.1 and 9.2

Exercise: Catalysis Mechanisms – 30 minutes

Now we will apply and play with these definitions.

1. Explain the key kinetic difference between specific and general catalysis. Use the example reactions below to illustrate your answer.³
 - (a) Write a mechanism and rate law for the specific base-catalyzed hydrolysis of an ester.
 - (b) Write a mechanism and rate law for the specific acid-catalyzed hydrolysis of an ester.
 - (c) Write a mechanism and rate law for the general base-catalyzed hydrolysis of an ester.
 - (d) Write a mechanism and rate law for the general acid-catalyzed hydrolysis of an ester.

³ Ch. 9.3

2. Explain the phenomenon of buffer catalysis in reactions that feature general acid or base catalysis.
3. A buffer is essential in experiments performed in water. Otherwise, we will not know the pH and cannot hold it to our chosen value. How can I find the rate of a reaction in the absence of buffer?
4. Describe a situation where the rate laws for general and specific catalysis mechanisms will be identical.
5. Explain why a reaction would be observed to proceed by a general catalysis mechanism rather than a specific catalysis mechanism?

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

1. Describe the various ways that a reaction can be catalyzed and understand how the reaction coordinate is changed with catalysis.
2. Be able to interpret the term “effective molarity” as it applies to intramolecular reactions and enzyme models.⁴
3. Construct a rate law for a catalyzed reaction.
4. Understand the difference between specific and general acid/base catalysis.

⁴ We will likely only mention effective molarity and other concepts of catalysis briefly in class. So little time is spent there. Thank goodness you have a great textbook to read. See chapter 9.2.

Looking Ahead

Acid and base catalysis serves as an important example for how kinetics can reveal mechanism. Next we will use this idea to interpret how rates of reaction in water change with pH. get ready for pH-rate profiles.